

Solar-powered BDD-electrolysis remediation of soil washing fluid spiked with diesel

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Abstract

BACKGROUND: The remediation of a soil spiked with diesel using Unitol L90 and sodium dodecyl sulfate (SDS) in a soil washing process was evaluated. The washing process was followed by an anodic oxidation based on boron doped-diamond electrolysis (BDD-electrolysis) to treat the washing liquid, using a solar photovoltaic (PV)-battery system as a direct electrical power source.

RESULTS: The results demonstrate that high concentrations of Unitol L 90 and SDS are required for the surfactant-aided soil-washing (SASW) to efficiently extract diesel from the soil. However, Unitol L 90 strongly interacts with the soil due to the adsorption process. The main characteristics of the effluents produced during this soil remediation was low conductivity. Therefore, the addition of 0.05 M Na₂SO₄ as an electrolyte, at current density of 30 mA cm⁻², was investigated. BDD-electrolysis demonstrated that total organic carbon (TOC) and total petroleum hydrocarbon (TPH) were removed more effectively using SDS. The key of the treatment is associated to the electrogeneration of S₂O₈²⁻ due to the contribution of the surfactant composition and the addition of 0.05 M Na₂SO₄. Additionally, an average solar irradiation intensity and integration of PV cells produced more than 10 A current and continuous electricity (~27 V), which was constantly supplied to the electrolytic reactor at all times through a DC generator.

CONCLUSIONS: Finally, based on the achieved results, BDD-electrolysis is an efficient technology approach to remove hydrocarbons from the effluent produced during the soil washing process.

Keywords: Soil washing; Surfactant; Total petroleum hydrocarbon; Boron-doped diamond; Renewable energy.

INTRODUCTION

During recent years, contamination of soils with organic compounds derived from hydrocarbons has gained attention due to the risk they pose to soil and water quality.^{1,2}

In this context, the petroleum industry refines crude oil into different compounds that can cause various carcinogenic and neurotoxic effects.³ Pollutants derived from petroleum can infiltrate the soil and groundwater in large quantities. If that occurs, these sites require remediation to reduce the risks associated with these hazardous materials.⁴

These contaminants are often associated with diesel, gasoline, fuel, oils, jet fuel, etc. Their impact to the environment is mostly due to their hydrophobic characteristics. Petroleum, for example, can be represented as total petroleum hydrocarbon (TPH), which is a broad family of organic compounds. These are sometimes referred to as gasoline and diesel range organics because their boiling point range are similar to those of gasoline (C_6 to C_{10-12}) and diesel (C_{8-12} to C_{24-26}), respectively.^{4,5}

Different processes can be applied to treat contaminated soils. Conventional treatment methods are based on chemical, physical, and biological techniques. Among the used technologies, soil washing is an efficient technique which is particularly

interesting due to the possibility to remove organic and inorganic compounds from the soil.⁶⁻⁹ Unfortunately, the high hydrophobicity of most TPH typically results in a low removal efficiency.¹⁰⁻¹⁴ Surfactant-aided soil-washing (SASW) is an ex-situ process which involves excavation of contaminated soil and its subsequent treatment at the surface.¹⁰ The surfactants used for SASW increase the solubility of hydrocarbons in water, improving the transport of contaminants in the soil.¹¹ Consequently, these surfactants affect the time necessary to treat a site compared to the use of water alone. Therefore, solubilizing agents such as Tween 80, β -cyclodextrin, Poloxamer 407, and sodium dodecyl sulfate (SDS), have been used for soil remediation.^{9,12-16}

In this context, López-Vizcaíno and coworkers removed phenanthrene from kaolinite clay using three types of surfactants: SDS, alkyl dimethyl benzyl ammonium chloride, and polyoxyethylene sorbitan monooleate (Tween 80)¹⁷. They observed that the anionic surfactant SDS achieved higher removal efficiency (90%) compared to the Tween 80 and the alkyl dimethyl benzyl ammonium chloride surfactants (70% and 30%, respectively). They concluded that the high adsorption of surfactants on the soil associated with nonionic and cationic surfactants could significantly affect the removal efficiency of phenanthrene.

Nevertheless, the SASW process produces an effluent, which consequently demands treatment. To treat this effluent, electrolysis with boron-doped diamond (BDD) electrodes has become a promising technology, since it can quickly remove all organic compounds, including surfactants.^{8,18-21} This behavior can be explained by the high oxidation power in the surface of the BDD anode. That oxidation power leads to a

competitive and parallel production of several oxidants, such as peroxodisulfates and active chlorine species (ClO^- , ClO_2^- , ClO_3^- , ClO_4^-).^{22,23}

Anodic oxidation with BDD anodes using direct current (DC) has been studied on the last decades. The used current can be derived from a conventional alternating current (AC) of an electrical grid by using a rectifier. The rectifier modifies AC into DC to feed the system.^{8,17–20,24–26} Therefore, the use of traditional electrical energy to power the BDD-electrolysis is a major challenge for the commercial feasibility of this technology. However, in the last decades due to availability, accessibility, and ease of use, solar photovoltaic (PV) panels are considered the most environmentally benign technology for energy production.^{26–29} In this context, the use of standalone/off-grid PVs to directly power electrochemical reactors has been investigated as a more eco-friendly technology for the remediation of contaminated wastewater. A solar panel plant consists of PV panels and batteries.^{26–28} Souza *et al.* (2015) have recently described the applicability of coupling PV power generators and electrochemical technologies to treat wastewater and soil containing pesticides. They observed that the mineralization efficiency of 2,4-D does not depend on the power source used. However, if solar PV panels directly power the electrochemical cell, fluctuations in the current intensity that affect the system are promoted.^{24,25}

The purpose of the present work is to evaluate the effectiveness of a treatment for soil spiked with diesel. The effects of nonionic (Unitol L 90) and anionic (SDS) surfactants used for SASW, which was followed by electrolysis with BDD anodes of the effluent, were compared. Moreover, the specific objectives of this study are to investigate: (i) the effects of surfactant concentration during the soil washing process,

but also on the characteristics of the produced washing waste; (ii) the influence of the surfactant on the elimination of organic compounds during BDD-electrolysis; and (iii) the effects of PV as a power source (solar PV-batteries system) on the performance of the BDD-electrolysis.

MATERIALS AND METHODS

Chemicals

In this study, diesel was selected as the soil contaminant. Unitol L 90 was selected for as the nonionic surfactant for the experiments. It was supplied by Oxitenio (Brazil). Its average ethoxylation degree is approximately 9, which reduces the molecule relative hydrophilicity, with a hydrophilic-lipophilic balance close to 16 (Table 1). SDS was used as an anionic surfactant (Panreac). Na_2SO_4 was supplied by Andriol (Brazil), hexane (Sigma Aldrich, Brazil).

Preparation of spiked soil

The soil used in this work was collected from the Paraná basin, Botucatu formation (PR, Brazil), where it was free from contamination. The main physical and chemical properties were measured according to the unified soil classification system (USCS) and standard methods of particle size distribution.^{30,31} The results are summarized in Table 2. The composition of the soil sample was approximately 95% of quartz, 3% of kaolinite, and 2% of calcite. The soil showed a higher percentage of Si (94.2%) in comparison to other soils with a high percentage of sand. The soil was air-dried and subsequently sieved (2 mm) to remove large stones and particles, and to form a fully saturated sample with an uniform distribution of contaminants. The diesel was dissolved

in hexane and thoroughly mixed with a measured amount of soil to obtain a final TPH concentration of 8951 mg kg^{-1} . This concentration was chosen to simulate an oil spill, as indicated in the literature.^{18,30,32} The spiked soil was aerated during two days to favor the evaporation the hexane.

Surfactant-aided soil-washing process

SASW with Unitol L90 or SDS surfactants as a fluid solution was carried out in a tank operated in batch mode.^{33,34} The tank volume was 1 dm^3 . Soil samples of 0.1 kg were uniformly spiked with diesel (10000 mg kg^{-1}) and subsequently mixed with 1 L of solubilizing agent (containing deionized water and different concentrations of Unitol L90 or SDS surfactants, ranging from 550 and 2500 mg L^{-1}) in the reactor for 6 h , at a stirring rate of 120 rpm . The same tank then acted as a settler for 24 h to separate the soil from the wastewater.

PV-battery system and soil washing treatment by BDD-electrolysis

For the BDD-electrolysis experiments, two coupled polycrystalline silicon solar PV modules (Canadian CS6U-325p) were connected in series with a total peak power of 640 W_p . As described in a previous work,³⁵ the PV modules were placed on the roof of the NUPER building, in the Federal University of Rio Grande do Norte/Natal, Brazil ($W 35^\circ 12'$, $S 05^\circ 54'$). The modules were installed facing south (20° W) and tilted at a 5° angle. Moreover, there was a MPPT charge controller (Victron Bluesolar 150/45-MC4) connected the PV modules, and two batteries (Solar Freedom, $12 \text{ V}/240 \text{ Ah}$) to control the produced electricity (Fig. 1). The BDD-electrolysis experiments were carried out in a laboratory-scale plant, in a single-compartment cell, as described in previous

works.^{33,34} BDD and Ti electrodes were used as anode and cathode, respectively. The characteristics of the BDD areas are as follows: 225 sp³/sp² ratio; 500 mg L⁻¹ boron content; and diamond layer of 2.68 μm width. To form the cell, a circular PVC was placed between the anode and the cathode plates, as seen in Fig. 1. The internal diameter and thickness of the PVC was 12 cm and 1.7 cm, respectively. The tests were performed using 0.9 L of soil washing effluent (with and without 0.05 M Na₂SO₄). The samples were placed in a thermos-regulated glass tank and circulated through the cell by a peristaltic pump at a 200 dm³ h⁻¹ flow. A constant applied current density (*j*) of 30 mA cm⁻² provided by a power supply (MPL-3505M, Minipa) was used throughout the BDD-electrolysis.¹³ However, before the BDD-electrolysis, the electrode was polarized for 10 min in a 0.05 M Na₂SO₄ solution at 30 mA cm⁻² to remove any impurities from its surface.² The batteries were directly connected to the Minipa triple power DC generator to regulate the current supply for the electrochemical system. Samples were withdrawn at a predetermined time interval to analyze chemical oxygen demand (COD), total organic carbon (TOC), persulfate, and TPH.

Analytical measurements

To determine the TPH concentration in the solid phase, a solid-liquid S-L extraction process was developed. First, 20 g of diesel-contaminated soil was extracted and transferred to a 40 mL glass centrifuge tube. Subsequently, 25 mL of chloroform was added to the tube, and the tube was closed. For the gas chromatography (Shimadzu) analysis, a Rxi-5ms column (5% diphenyl/dimethylpolysiloxane) of 30 m was used with

an injector temperature of 90 °C and transfer temperature of 120 °C. The percentage of TPH removal was estimated using the following equation:

$$TPH \text{ removal } \% = \left(\frac{TPH_0 - TPH_f}{TPH_0} \right) * 100 \quad (1)$$

where TPH_0 and TPH_f are the total petroleum hydrocarbon at time $t = 0$ (initial) and at f (time), respectively. The critical micelle concentration (CMC) was determined by measuring the surface tension at a fixed temperature (25 ± 1 °C) using a Sensa Dyne Surface Tensiometer.³⁶ The variation in surface tension as a function of Unitol L90 and SDS concentration was determined in the presence of 0.05 M Na_2SO_4 , and 3% of SDS or Unitol L90, in 30 mL. CMC is represented by the break in the plots of concentration versus surface tension. To determine the pH and conductivity of the soil samples, the standard method EPA 9045C was used. This method involves the mixture of 10 g of dry soil with 25 mL of distilled water, which are subsequently magnetically agitated for 10 min. After the elimination of contaminants from the soil washing effluent by the BDD-electrolysis, the dissolved organic carbon was measured using an Analytik Jena TOC analyzer. The non-purgeable organic carbon method was used to generate reproducible TOC values with an accuracy of $\pm 2\%$. The COD levels were measured by using pre-dosed reagents (HANNA® vials) in 2 mL samples. Samples for COD determination were digested in a thermal reactor (HANNA instrument) at 150 °C for 2 h. Subsequently, the samples were analyzed on a spectrophotometer (Hanna HI 83099), at 25 °C. Three measurements were taken, and the average value with a standard deviation of $< 3\%$ was recorded. The persulfate produced during BDD-electrolysis was monitored. Aliquots of 200 μ L were analyzed by using an in situ chemical oxidation method³⁷ from 190 to 600 nm, with a Shimadzu spectrophotometer, model 1800.

The solar irradiation and charge of both batteries by the electricity from the PV panels were monitored by a solar radiation meter (PCE-SPM 1) and a power multimeter (D52-2047), with sensors mounted between the PV cell and the MPPT, and between the MPPT and the batteries, respectively. Additionally, the current efficiency (CE) for each treatment was estimated according following equation:³⁸

$$CE (\%) = \left(\frac{(\Delta COD)_{exp} F V_s}{8 I \Delta t} \right) \times 100 \quad (2)$$

where $(\Delta COD)_{exp}$ is the experimental COD decay, F is the Faraday constant (96487 C mol^{-1}), V_s is the solution volume (L), 8 is a conversion factor, I is the applied current (A), and t is the electrolysis duration in s.

The energy consumption (EC) per unit volume (kWh L^{-1}) during the BDD-electrolysis of soil washing fluid can be calculated from equation 3:

$$EC = \left(\frac{E_{cell} I t}{(\Delta COD) V_s} \right) \quad (3)$$

where E_{cell} is the average cell potential (in V), I is the applied current (A), t is the duration of the BDD-electrolysis (h), and COD is decay in COD .³⁸

RESULTS AND DISCUSSION

SASW process

The soil washing process using Unitol L90 and SDS surfactants has been used as an ex-situ process that improves the removal of soil contaminants.^{12,33} As the solubility of diesel in water is low, it is necessary to add a surfactant to aid the production of micelles, improving the pollutant extraction. The removal efficiency of TPH using SDS is higher compared to the one using Unitol L90, for all concentrations (Fig. 2). The TPH

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removal efficiency during the SASW process by SDS reached approximately 83%, and by Unitol L90 approximately 74%, both at a surfactant concentration of 2500 mg L⁻¹. Lower surfactant concentrations led to lower removal rates, 52% and 67% by Unitol L90 and SDS, respectively, at 550 mg L⁻¹. This behavior can be explained by the fact that at lower concentrations, surfactant molecules can be presented as dispersed monomers.^{39,40} This suggests that CMC is an important information for the SAWS process, because the formation of micelles could contribute to the removal of organic compounds. For this reason, the CMC of Unitol L90 and SDS in the water-diesel effluent, at different concentrations and surface tension, were determined (Fig. 3(a,b)). The obtained CMC result for Unitol L90 was 641 mg L⁻¹, and 1621 mg L⁻¹ for SDS. Therefore, nonionic surfactants have low CMC values, leading to an easier micellization, since aggregation occurs mainly with hydrophilic chains.^{39,40} Nevertheless, the solubilization of contaminants occurs mainly when the concentration of surfactant in the soil washing increases, favoring the mobilization and solubilization of hydrocarbons in the aqueous media. As observed, SDS removed more effectively the TPH from the soil because anionic surfactants have lower adsorption in the soil, which substantially increases availability for micelle formation.^{14,41}

Influence of surfactant on effluent treatment by BDD-electrolysis

As described in a previous work,³⁵ the weather in Natal is very sunny nearly every day owing to the proximity of the city to the equator. It was observed that it is possible for two PV modules to generate current intensities of over 10A for 8-9 h/day when the irradiation intensity is greater than 15000 kJ m⁻², with an almost constant voltage of ~27

V recorded from the PV to the batteries. As seen in Fig. 4(a,b), the normalized COD decreases as a function of time in the absence and presence of 0.05 M Na₂SO₄ at different concentrations of Unitol L90 and SDS, respectively. The fluid produced during SASW presented low conductivity (approximately 489-536 μS cm⁻¹ and 656-793 μS cm⁻¹ in the presence of Unitol L90 and SDS, respectively). No significant COD decrease was obtained without a supporting electrolyte in the effluent. This result indicates a limited COD removal via direct electrooxidation by strong oxidants that produce (•OH). In contrast, in the presence of 0.05 M Na₂SO₄, the efficiency of organic compounds removal increased for all effluents. This behavior can be related to the production of •OH on the BDD surface by direct electrooxidation and indirect oxidation in the presence of sulfates. These sulfates contribute to the production of S₂O₈²⁻ from SO₄²⁻, as shown in equations 4-6.^{38,42,43}



SDS removed 83% of TPH from the soil after the soil washing process, whereas Unitol L90 removed 74%. In addition, a significant COD removal (approximately 82%, 89%, and 94%) was obtained by using SDS in the presence of supporting electrolytes (Fig. 4(a)). Whereas in the presence of Unitol L90 and 0.05 M Na₂SO₄, the COD removal efficiency was lower (68%, 75%, and 83%). The increased removal efficiency by SDS can be attributed to the sulfate released from SDS fragmentation. The sulfate is electrochemically converted to persulfate, providing a synergetic effect due to the

electrooxidation of SO_4^{2-} into $\text{S}_2\text{O}_8^{2-}$. Moreover, the oxidation of TPH in the presence of Unitol L90 can be influenced by the steric hindrance phenomenon at the micelles and the spatial configuration of micelles, which reduces the availability of hydrophobic tail of surfactants towards oxidant species.⁴⁴⁻⁴⁷ An easier micellization occurs because of the aggregation due to hydrophobic attraction among nonpolar chains, which leads to a higher proportion of micelles in the effluent.³⁹

TOC and TPH removal from the soil

The high toxicity of TPH is associated with a broad family of organic compounds, represented as a function of equivalent carbon number. Therefore, Brazilian environmental laws are strongly restrictive regarding the release of TPH in aquatic environments and in the soil.⁴⁸ Fig. 5(a) shows the percentage of TOC removal from different effluents by a BDD-electrolysis of 480 min, at 30 mA cm^{-2} . The removal is a function of Unitol L90 and SDS concentration. Based on the experimental conditions, the wastewater generated from SASW presents a high amount of organic matter (approximately 1978 mg L^{-1} and 2433 mg L^{-1} in the presence of Unitol L90 and SDS, respectively). Interestingly, in the anodic oxidation of effluents using SDS, it is possible to observe an increased efficiency that can be explained by the oxidation of SO_4^{2-} into $\text{S}_2\text{O}_8^{2-}$.⁴⁴ Moreover, the TOC removal in the presence of Unitol L90 at different concentrations reached 56%, 62%, and 75% (Fig. 5(a)). The TPH removal by both surfactants is illustrated in Fig. 5(b). Increasing concentration of surfactants favored the removal of TPH. After 480 min of electrolysis, TPH removal reached approximately 78%, 82%, and 89% by SDS, and 59%, 65%, and 71% by Unitol L90. Similar results

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have been recorded in a previous work,⁴⁶ where it was observed that SDS encapsulates or concentrates the organic compounds by forming micelles. It was also observed that during indirect electrochemical oxidation, the micelles are degraded by hydroxyl radicals, and sulfate is released from the fragmentation of SDS, which is converted into persulfate.

Electrochemical production of persulfate

Fig. 6(a,b) presents the persulfate production during BDD-electrolysis at 30 mA cm⁻² in 0.05 M of Na₂SO₄ according to the concentration and type of surfactant. Lower concentrations of S₂O₈²⁻ were produced in the presence of Unitol L90 (Fig. 6(a)), whereas in the presence of SDS, 0.90, 1.10, and 1.72 mM of S₂O₈²⁻ were produced after 480 min (Fig. 6(b)). The results indicate that S₂O₈²⁻ production depends on the concentration and type of surfactant in the effluent. Considering the composition of the soil washing fluid, a production of S₂O₈²⁻ derived from SO₄²⁻ decomposition is expected in the reaction medium.⁴⁶ Similarly, previous studies reported that during remediation involving SDS and electrochemical treatments, SO₄²⁻ ions can be produced and subsequently oxidized into persulfate by BDD electrodes.^{43,44,49} Moreover, considering the easier micellization of nonionic surfactants, the effluent containing Unitol L90 presented a lower persulfate production, which can be justified by the steric hindrance.

Economic analysis

Powering the BDD-electrolysis system with green energy is a challenge for the development of this novel eco-friendly environmental remediation technology. In this

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context, a cost analysis was performed, and the results were compared with a system powered by a conventional source of energy. Fig. 7(a) shows the CE (%) in the presence of Unitol L90 and of Na₂SO₄ during BDD-electrolysis at 30 mA cm⁻². It can be observed that CE increases with increasing surfactant concentration, which indicates maximum interaction between the hydroxyl radicals and the contaminants. A similar behavior was observed in the presence of SDS, but with a higher CE percentage (Fig. 7(b)). However, after the first 30 min, the CE decreases due to the promotion of secondary reactions, such as the production of oxygen and persulfate.⁴⁴ EC is an important parameter to estimate the amount of PV cells on an industrial scale and to assess the feasibility to shift from conventional to PV-based treatment. Fig. 7(c) shows the EC according to COD removal for different concentrations of Unitol L90 and SDS, at 30 mA cm⁻². These values were calculated using equation 3. Fig. 7(c) shows that in lower concentrations, both surfactants led to lower EC, especially SDS. In this context, it is important to consider that micellization is easier for nonionic surfactants because aggregation occurs mainly due to the hydrophobic attraction among nonpolar chains, consequently, more micelles are available. Moreover, nonionic surfactants can be adsorbed from the soil, contributing to the partitioning of hydrophobic contaminants into surfactants hemimicelles in the soil surface, which in this case, was at a concentration mostly below CMC.⁴⁰ The results indicate that the integration of PV and BDD-electrolysis as a sustainable system could be used on an industrial scale and in places that do not have easy access to conventional energy.

CONCLUSIONS

The most important conclusions of this work can be summarized as follows:

- SASW was considered an efficient technology to remove diesel from soils. At a 2.5 g kg^{-1} surfactant concentration in the soil, Unitol L90 and SDS led to a 74% and 84% TPH reduction, respectively. The characteristics of the effluent produced by SASW depend on the dose of surfactant used. The highest pollutant removal was obtained with the highest SDS concentration.

- BDD-electrolysis of the wastewater produced by soil washing is efficient in the removal of TPH and TOC. The most efficient removal was in presence of 0.05 M Na_2SO_4 and with the highest SDS dose.

- Persulfates are produced during BDD-electrolysis in both effluents (with Unitol L90 and SDS) produced by SASW. However, when SDS was used as the soil washing fluid, there was a more effective increase in the $\text{S}_2\text{O}_8^{2-}$ concentration due to the oxidation of sulfates. The efficiency of anodic oxidation in the removal of TPH depends on the concentration and type of surfactants in the effluent.

- BDD-electrolysis can be powered using a solar PV-battery system for the treatment of soil washing fluid from SASW. The solar radiation in Natal contributes to the application of a solar PV-battery system. Finally, the features of the direct connection between BDD-electrolysis and PV systems should be explored. Therefore, further experiments are in progress to improve the operation process of directly connected PV systems.

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Table 1. Properties of surfactants.

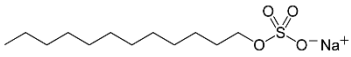
Surfactant	Type	Formula	MW	CMC (mg L ⁻¹)	Molecular structure
SDS	Anionic	C ₁₂ H ₂₅ NaO ₄ S	288.37	1621	
Unitol L 90	Non-ionic	C ₃₀ H ₆₂ O ₁₀	582.82	641	

Table 2. Properties of soil.

Chemical composition (weight %)	
Si	94.28
Al	3.41

Fe	1.82
Ti	0.17
Ca	0.08
K	0.07
S	0.06
Mg	0.06
P	0.04
Mineralogical composition (weight %)	
Quartz	95
Kaolinite	3
Calcite	2
Other properties	
Coefficient of non-uniformity (CNU)	2.39
Coefficient of curvature (CC)	1.01
Plasticity index (IP)	non-plastic
pH	5.43
Conductivity ($\mu\text{S cm}^{-1}$)	10.11
TPH (mg kg^{-1})	8951

Figure captions

Figure 1.

Schematic of electrochemical flow cell integrate with PV-Battery system powered of soil washing fluid at BDD anodes. (1) Anodic part, (2) BDD/Si, (3) silicon seal, (4) electrolytic compartment, (5) silicon real, (6) Ti electrode and (7) cathode part.

Figure 2.

(a) Elimination of TPH load as a function of surfactant/soil ratio. (■) Unitol L90 and (●) SDS. Inset: TPH removal rate after 6 h.

Figure 3.

Effect of the superficial tension as a function of surfactant concentration. Experimental conditions: water-diesel effluent in Na_2SO_4 (0.05 M). (a) Unitol L90 and (b) SDS.

Figure 4.

COD decay during the BDD-electrolysis treatment of the soil washing effluents in the absence and in the presence of Na_2SO_4 0.05 M. (a) Unitol L90 and (b) SDS. (♦) 550 mg L^{-1} , (●) 1000 mg L^{-1} and (■) 2500 mg L^{-1} . Without Na_2SO_4 (empty symbols) (○) 550 mg L^{-1} and (

Figure 5.

Effect of the surfactants concentration on TPH removal efficiency during the BDD-electrolysis treatment in 0.05 M Na_2SO_4 using BDD anode and Ti cathode at 8 h electrolysis time. (a) TOC removal and (b) TPH.

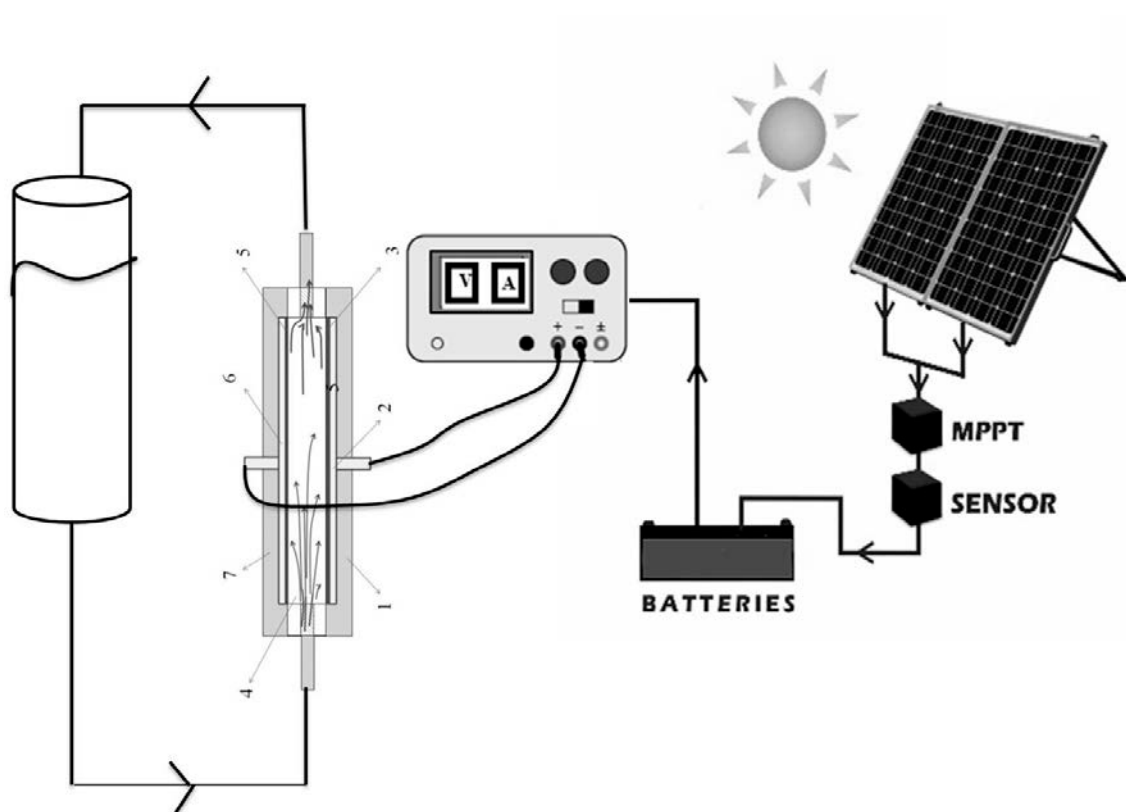
Figure 6.

Electrochemical generation of persulfate as a function of effluents obtaining during SASW process in presence supporting electrolyte (a) Unitol L90 and (b) SDS. (♦) 550 mg L⁻¹, (●) 1000 mg L⁻¹ and (■) 2500 mg L⁻¹.

Figure 7.

Energy consumption and current efficiency (CE, in %) versus concentrations of surfactants at the experimental conditions: 30 mA cm⁻² and 25°C. (a) Unitol L90 and (b) SDS.

Figure 1



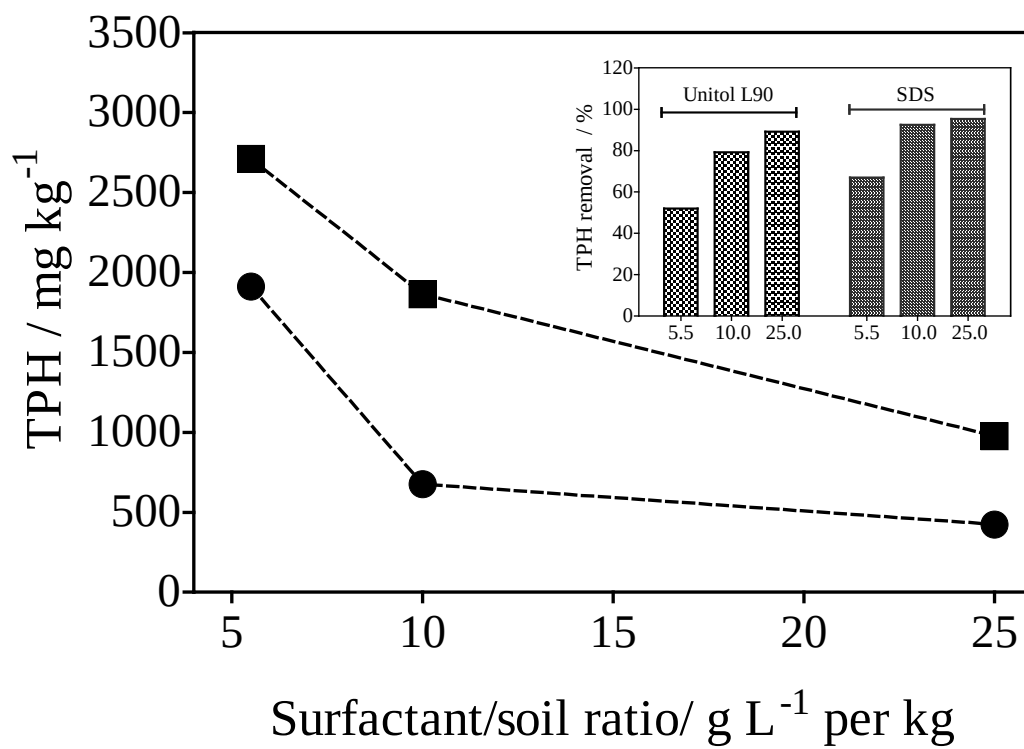
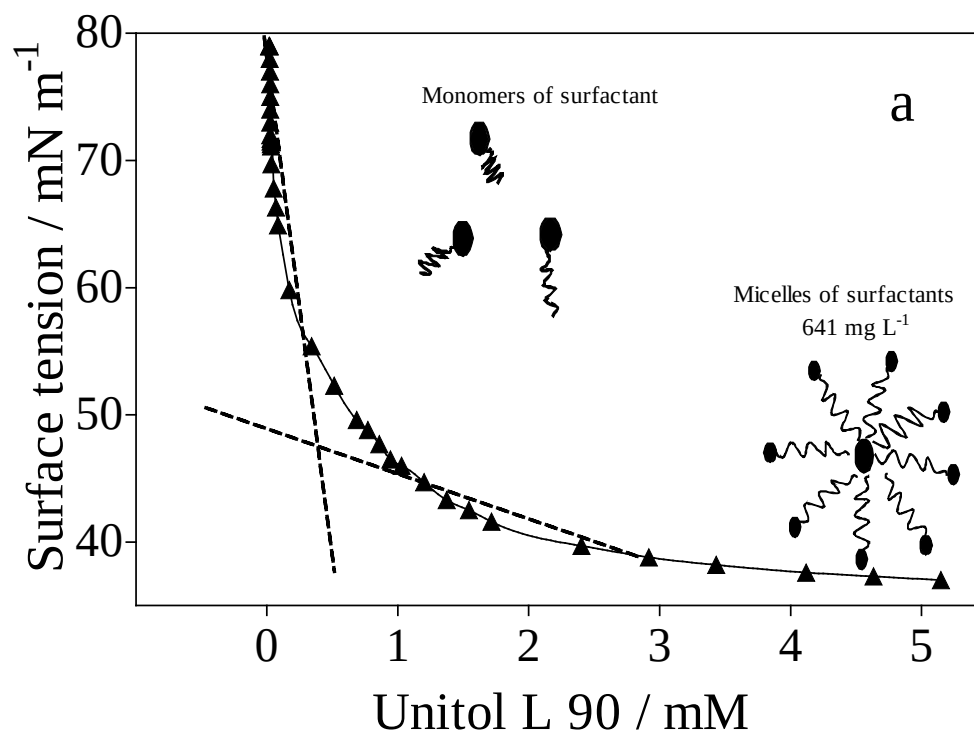


Figure 2

Figure 3

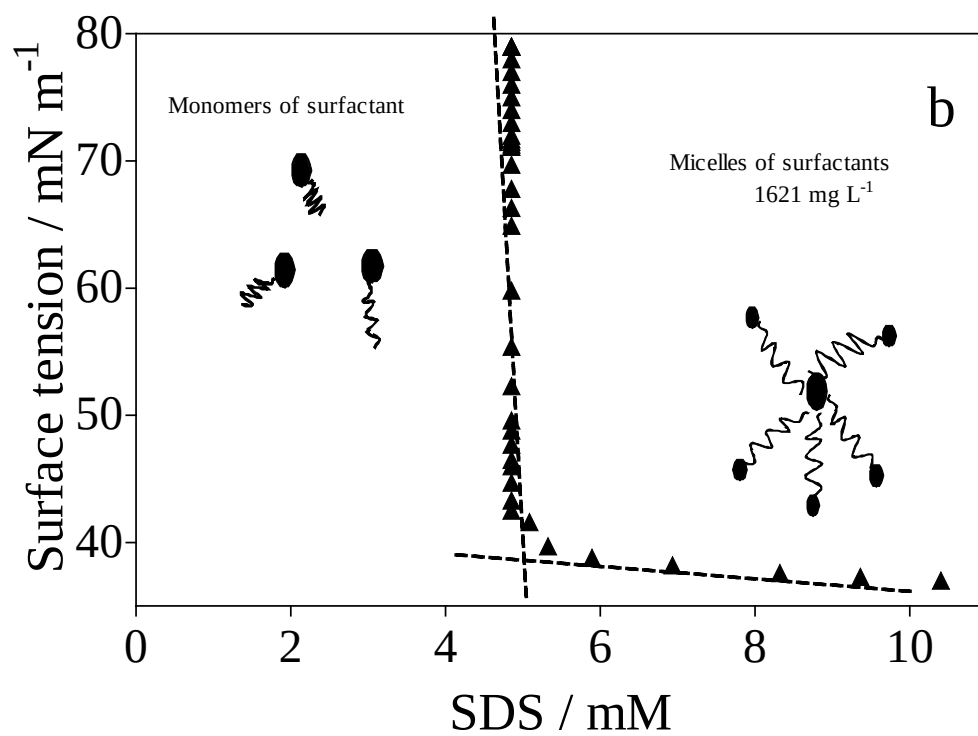


Figure 4

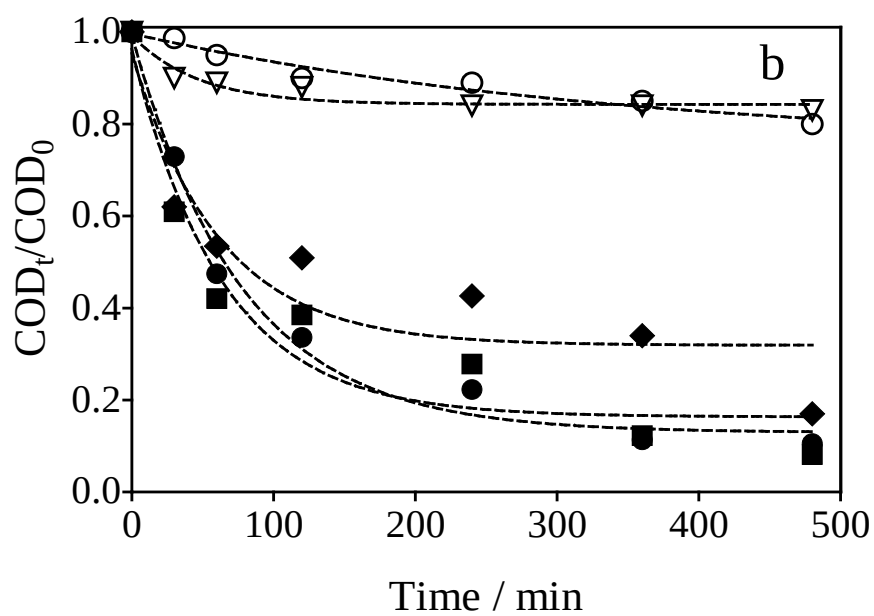
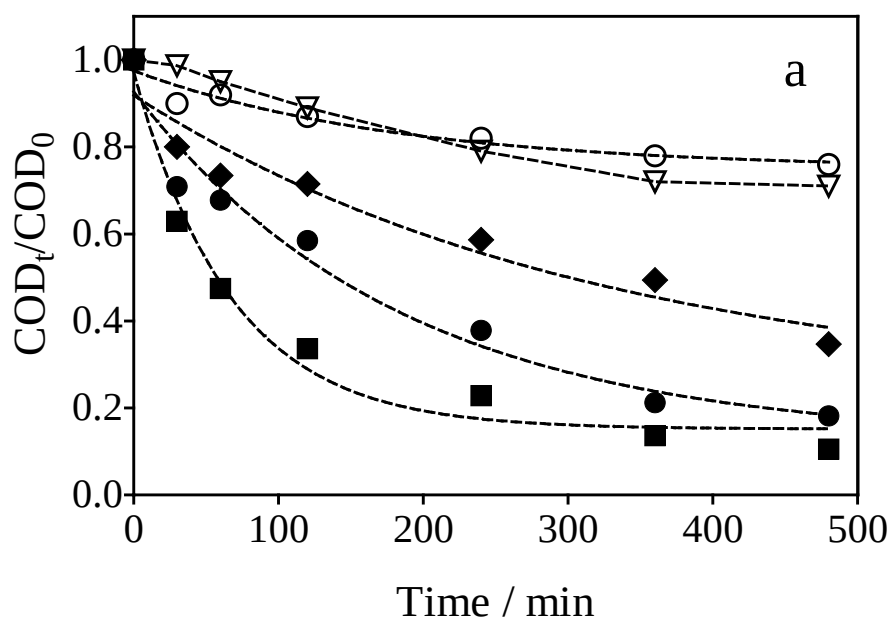
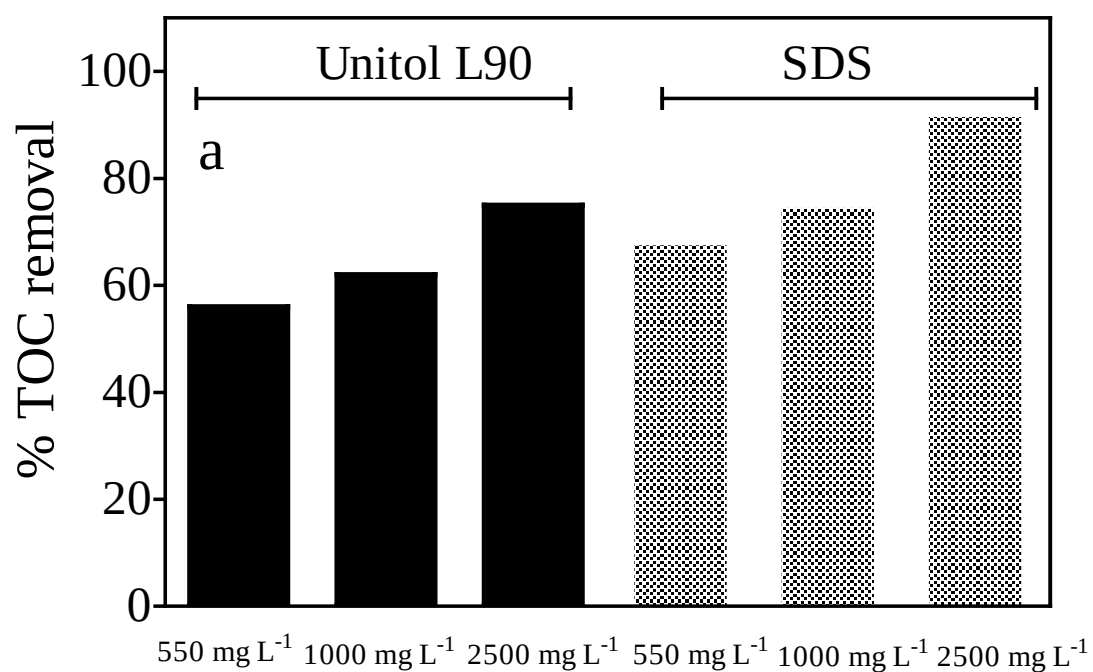


Figure 5

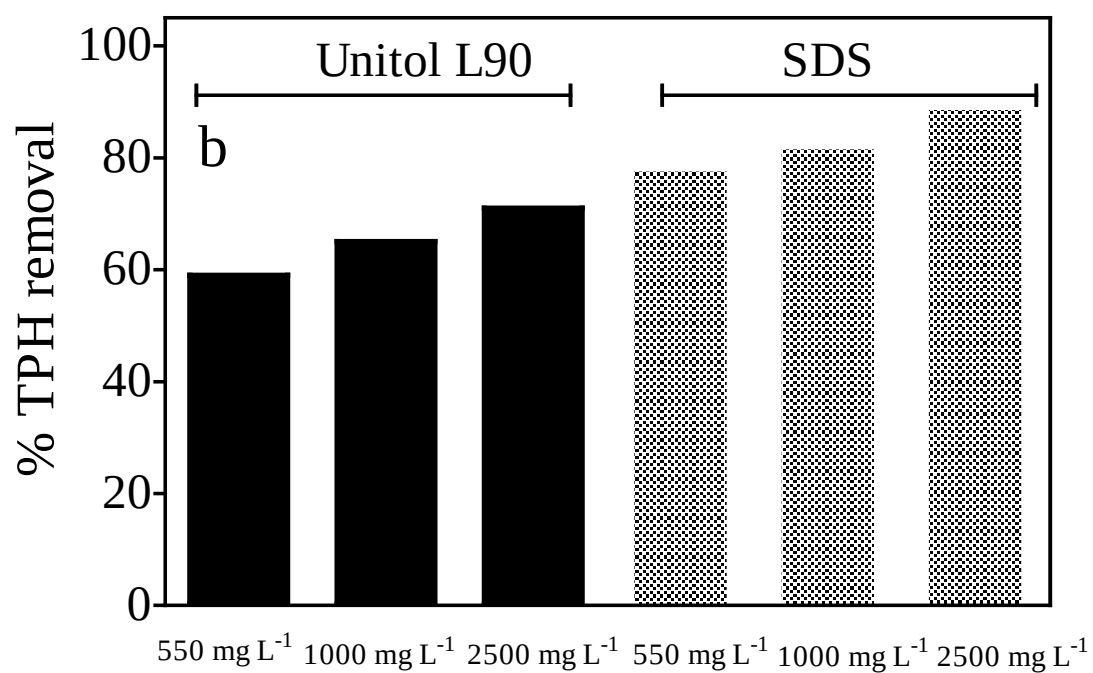


Figure 6

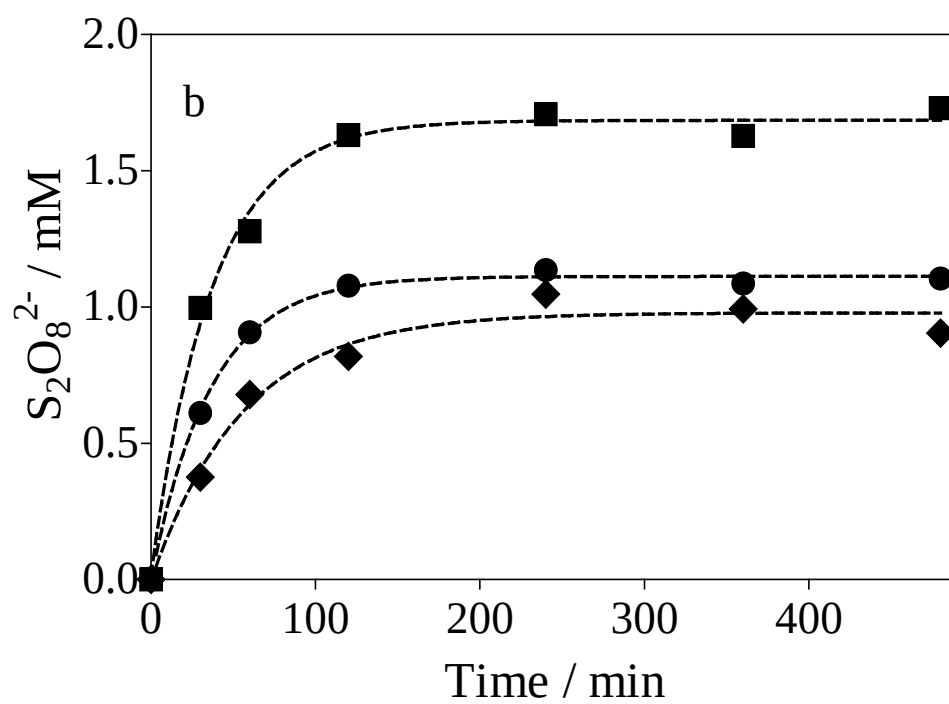
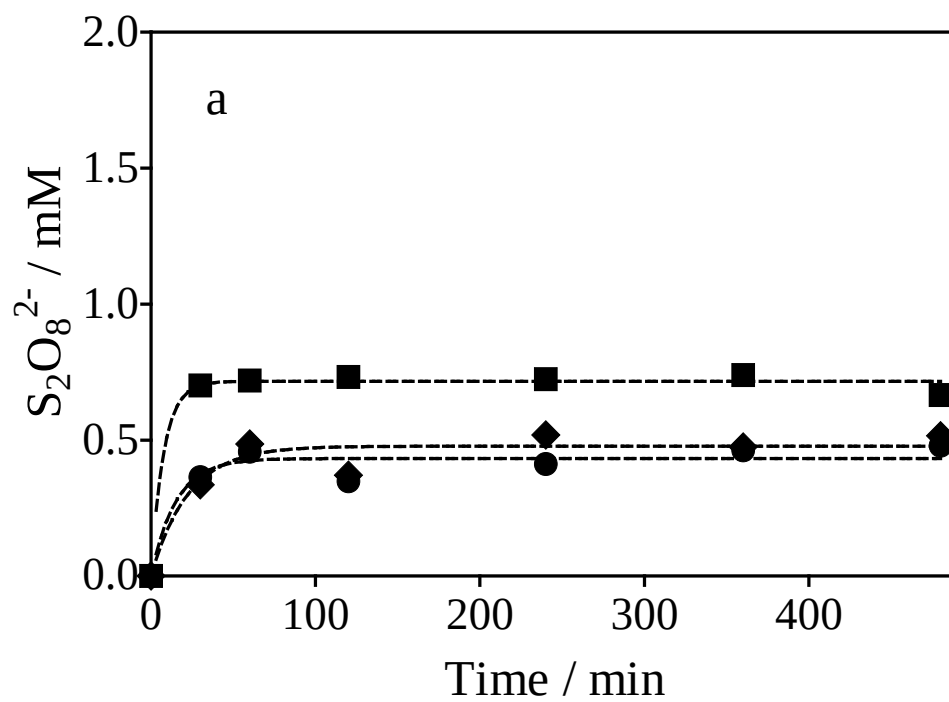


Figure 7

